[Physical World Modeling for Embodied Manipulation]

Workshop https://phiwm.github.io/CORL2026.html

Organizers

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Our group is composed of a diverse set of organizers across career stages, sectors, institutions, and geographies. The organizing team includes PhD students, a postdoctoral researcher, assistant professors, full professors, and one industry research scientist. The team also reflects diversity, and broad international representation, spanning seven countries across three continents: Asia, Europe, and North America. At least one organizer will attend CoRL 2026 in person; Runyi Yang will serve as the corresponding organizer and will attend the required workshop organizers meeting. Additional organizers are expected to attend in person if the workshop is accepted.

Description

World models for embodied agents are rapidly emerging from several active research streams.
Video-based world models focus on predicting or generating plausible future observations from visual inputs, often emphasizing temporal consistency, controllability, and long-horizon visual imagination [1, 3]. **3D-based world models** focus on persistent geometry, spatial structure, scene layout, viewpoint consistency, and object- or scene-level representations that support spatial reasoning [2, 4]. **Robot world models** are typically built upon 3D- or video-based world modeling frameworks. While leveraging the strengths of these foundations, they place particular emphasis on action-conditioned prediction, planning, policy evaluation, and control under embodiment-specific constraints, with the goal of bridging perception and executable robot behavior [5].
Robot manipulation is also becoming increasingly challenging. New platforms such as dexterous hands, mobile manipulators, and humanoids make richer interaction possible, while manipulation tasks increasingly involve contact-rich behavior, tool use, and long-horizon reasoning. In these settings, action decisions depend on physical state that may not be explicit in images or geometry alone. Cutting food, pouring liquid, or moving a heavy object are not only problems of recognizing shape and pose; they require estimating how the world will respond to contact, force, and motion. This raises a central question for robot learning: what should a world model represent when an embodied agent must not only observe its environment, but physically interact with it?
Thus, we propose this workshop focuses on **physical world modeling** as a key next step toward embodied intelligence.
A **physical world model** should go beyond visual appearance, geometric reconstruction, or task-specific policy prediction. It should represent physically actionable state and predict the conse-

quences of interventions: what objects are present, how they are arranged, how they can move, how they interact under contact, which actions are feasible, and which future states may result. For real-world robot actions, this requires reasoning about contact, force, friction, support relations, stability, articulation, deformability, material properties, affordances, uncertainty, and multimodal feedback from vision, depth, touch, force/torque, proprioception, audio, and other sensors. The goal is not only to generate visual observations, but to build models that can support reliable physical observations and actions.

Despite rapid progress, physical world modeling remains at an early stage. Researchers are currently drawing ideas from video generation, 3D/4D scene representation, differentiable simulation, real-to-sim and sim-to-real transfer, inverse physical inference, multimodal foundation models, and vision–language–action systems. However, these communities often use different assumptions, representations, datasets, and evaluation protocols. Video models may capture rich temporal priors but lack explicit physical state; 3D models may provide spatial structure but not interaction dynamics; robot models may support control but remain limited by embodiment-specific data and weak generalization.

A dedicated workshop is therefore needed to clarify how these threads should be connected for physically grounded embodied intelligence. This half-day in-person workshop will bring together researchers in robot learning, manipulation, computer vision, 3D/4D reconstruction, graphics, simulation, physical reasoning, multimodal learning, and foundation models.

Core Challenges / Research Questions

The workshop will center on three questions:

Representation: what is a physical world model? Physical world models should go beyond visual observation, video prediction, or pure 3D geometry to encode the physical variables that determine how the world can be acted upon and how it will change under action. This includes multimodal signals such as RGB, depth, tactile feedback, force/torque, proprioception, thermal cues, and audio, as well as object-level physical properties such as pose, shape, material, mass, friction, compliance, articulation, contact state, support relations, stability, affordances, and uncertainty. The goal is to move from models that merely represent how the world appears to models that expose physically actionable states for prediction, planning, control, adaptation, and real-world execution.

Construction: how do we build such models at scale? Modern robots observe the world through sparse RGB, RGB-D or wrist cameras, human videos, robot logs, and occasionally simulation assets. The workshop will bring together methods for scalable 3D reconstruction, 3D Gaussian scene encoding, differentiable digital twins, real-to-sim-to-real pipelines, and multimodal pretraining. We will discuss how to convert unstructured physical observations into robot-usable state representations without requiring a fully manually authored simulator for every scene.

Invited Speakers

Saining Xie REQUESTED
AMI Labs & NYU `saining.xie@nyu.edu`

World models for embodied intelligence: toward systems grounded in continuous sensory experience and real-world interaction.

Ming-Yu Liu COMMITTED
NVIDIA `mingyul@nvidia.com`

Omnimodal World Models for Physical AI

Tamim Asfour COMMITTED
KIT asfour@kit.edu
Developing humanoid systems that combine perception, manipulation, learning, and human-robot interaction to operate in complex real-world environments.

Manling Li COMMITTED
Northwest University manling.li@northwestern.edu
Multimodal embodied agents: building systems that connect language, vision, and action to support grounded reasoning, physical-world understanding, and interactive planning.

Proposed Format and Interactivity

The workshop will be a half-day in-person event combining invited talks, contributed presentations, posters, live demos, and community discussion. We will accept only **non-archival submissions**, including 4-page papers: position papers, work-in-progress papers, negative results, and 8-page papers discussing impressive recent progress or strong preliminary results. This format encourages the open discussion of emerging ideas without preventing future archival publication.

The program will include brief organizer remarks, four 30-minute keynote talks, and two contributed tracks: a **poster track** and a **demo track**. Each track will select three outstanding submissions for short oral presentations. The poster and demo sessions will run in parallel, allowing participants to engage directly with authors, inspect interactive systems, ask technical questions, and compare different approaches to physical world modeling.

The demo track will encourage physical robot demos, executable world-model prototypes, multimodal sensing systems, manipulation benchmarks, real-to-sim-to-real examples, and interactive visualizations. The workshop will also include a coffee break for informal exchange and a moderated panel discussion on what physical world models should encode, how they should be constructed from multimodal real-world data, and how they should be evaluated through executable robot behavior. The workshop will conclude by recognizing one **Best Paper** and one **Best Demo**.

Contributed Papers Plan

We will solicit **non-archival workshop submissions** through the workshop website, mailing lists, relevant research communities, and social media. Submissions may include short papers, position papers, work-in-progress papers, negative results, benchmark or dataset proposals, system descriptions, and papers showing impressive recent progress or strong preliminary results.

The workshop will include a **poster track** and a **demo track**. The poster track will focus on physical state representation, multimodal sensing, action-conditioned prediction, robot manipulation, simulation-to-real transfer, evaluation protocols, and embodied datasets. The demo track will invite interactive systems such as robot demonstrations, executable world-model prototypes, multimodal sensing platforms, manipulation benchmarks, and visualization tools.

All submissions will be reviewed by the organizers and a program committee with expertise in robotics, vision, learning, simulation, and embodied AI. Reviews will consider relevance, technical quality, clarity, novelty, potential impact, and suitability for an interactive half-day workshop. Three submissions from each track will be selected for short oral presentations. The workshop will also recognize one **Best Paper** and one **Best Demo**.

Plans to Encourage Participation and Diversity

We will encourage participation beyond the invited speaker list through an open call for **non-archival submissions**, including short papers, position papers, work-in-progress results, negative results, benchmark proposals, and demo submissions. The poster and demo tracks are designed to broaden participation beyond traditional paper presentations, giving students, early-career researchers, system builders, and interdisciplinary contributors multiple ways to engage with the workshop. Selected submissions from both tracks will receive oral presentations, and the poster/demo session and panel discussion will provide additional opportunities for direct interaction with speakers, organizers, and attendees.

Our organizing team is itself diverse across career stages, sectors, institutions, and geographies. It includes PhD students, a postdoctoral researcher, assistant professors, full professors, and an industry research scientist, with broad international representation across seven countries and three continents: Asia, Europe, and North America. We will use this network to disseminate the call broadly across robotics, computer vision, machine learning, simulation, and embodied AI communities.

We will also actively promote diversity in the workshop program. In selecting speakers, reviewers, oral presentations, posters, demos, panelists, and awardees, we will aim for balanced representation across career stage, gender, geography, institution type, research background, and sector. We will avoid overrepresentation from any single institution or region, encourage submissions from junior researchers and underrepresented communities, and ensure that the panel and discussion format gives meaningful opportunities to contribute to other participants than the invited speakers .

Logistics and Resources

Estimated audience size: We expect approximately 80–120 in-person participants. If virtual participation or livestreaming is supported by the conference, we expect an additional 50–100 remote attendees.

Physical requirements: We request a standard workshop room with projector, screen, microphones, speakers, and reliable Wi-Fi. For the poster and demo session, we request poster boards for accepted poster-track papers and several tables for demo-track presentations. Demo tables should ideally have nearby power outlets or power strips. If space allows, a small open area would be useful for lightweight robot or interactive system demonstrations.

Special requirements: No major special requirements are needed beyond standard Audio-Visual support, poster boards, demo tables, power access, and Wi-Fi. For safety and logistics, any physical robot demos will be limited to compact, tabletop, or otherwise conference-safe setups.

Commitments

✓ At least one organizer will attend CoRL 2026 in person.
✓ Organizers will attend the CoRL 2026 workshop organizers meeting.
✓ Organizers will submit a post-workshop artifact.
✓ Willingness to merge with related proposals if requested.

References

[1] Dacheng Li, Yunhao Fang, Yukang Chen, Shuo Yang, Shiyi Cao, Justin Wong, Michael Luo, Xiaolong Wang, Hongxu Yin, Joseph Gonzalez, et al. Worldmodelbench: Judging video generation models as world models. *NeurIPS*, 2026.

[2] Sikuang Li, Chen Yang, Jiemin Fang, Taoran Yi, Jia Lu, Jiazhong Cen, Lingxi Xie, Wei Shen, and Qi Tian. Worldgrow: Generating infinite 3d world. In *AAAI*, 2026.

[3] Hao Liu, Wilson Yan, Matei Zaharia, and Pieter Abbeel. World model on million-length video and language with blockwise ringattention. In *ICLR*, 2025.

[4] Chaojun Ni, Xiaofeng Wang, Zheng Zhu, Weijie Wang, Haoyun Li, Guosheng Zhao, Jie Li, Wenkang Qin, Guan Huang, and Wenjun Mei. Wonderturbo: Generating interactive 3d world in 0.72 seconds. In *CVPR*, 2025.

[5] Yu Shang, Xin Zhang, Yinzhou Tang, Lei Jin, Chen Gao, Wei Wu, and Yong Li. Roboscape: Physics-informed embodied world model. *NeurIPS*, 2026.